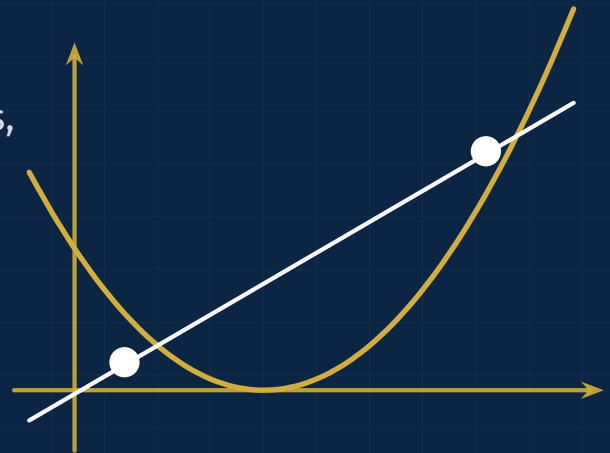


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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 10

Differentiation

$\frac{dy}{dx}$ is the gradient of the curve at any point. Multiply by the power, take one off — then use it for gradients, tangents, turning points, best boxes and moving particles.

**5**

VARIANTS

29

PAST-PAPER PARTS

22

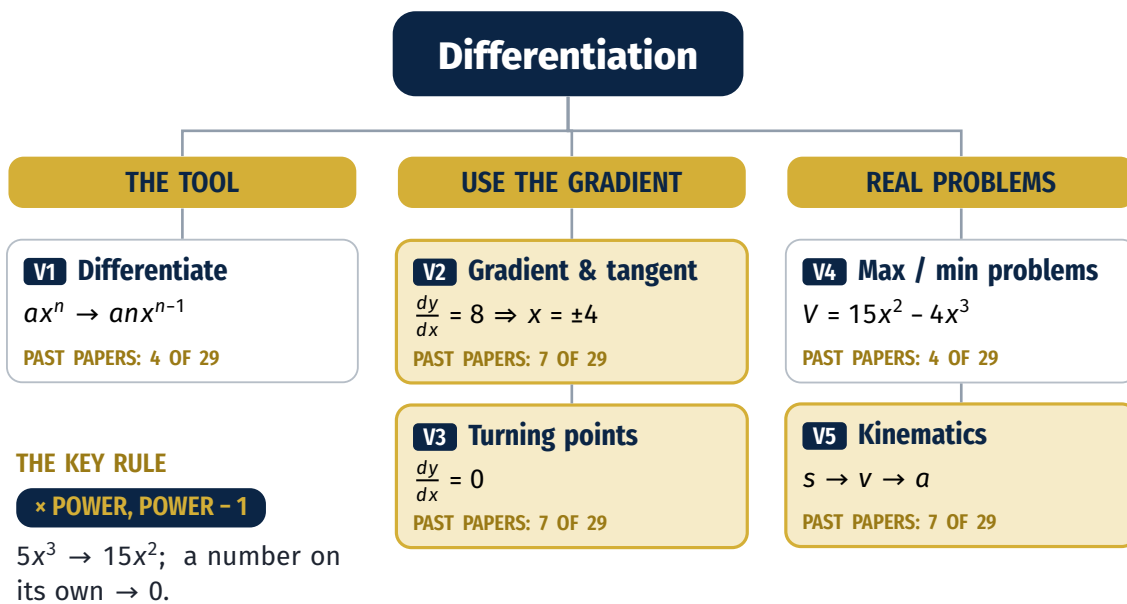
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Differentiate **V2** Gradient & tangent **V3** Turning points**V4** Maximum & minimum problems **V5** Kinematics: s , v , a

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Differentiation at a glance



Your notes


★ How often does each variant come up?

23 past-paper questions (Jan 2020 → Nov 2025) hold 29 parts; 22 practice parts — grouped by type.

	PAST PAPERS (OUT OF 29)	PRACTICE PARTS
V1 Differentiate	4	3
V2 Gradient, tangent	7	3
V3 Turning points	7	9
V4 Max, min problems	4	4
V5 Kinematics	7	3

WHAT THIS TELLS YOU On every paper, near the end (Q11–24, 4–8 marks). The differentiating is easy; the marks are for what you do next: set $\frac{dy}{dx}$ equal to something, solve, then substitute back.

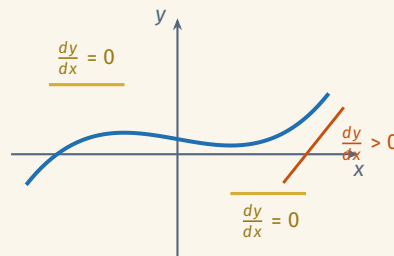
BEFORE YOU START

The big idea

$\frac{dy}{dx}$ = the gradient at any x .

For a curve the gradient keeps changing. Differentiating gives a **formula for the gradient**: put in x , get the gradient there.

Where $\frac{dy}{dx} = 0$ the curve is **flat**: a turning point.



Your notes



The rules

Power

$$ax^n \rightarrow anx^{n-1}$$

× power, power -1.

x term

$$7x \rightarrow 7$$

The x disappears.

Number

$$5 \rightarrow 0$$

A constant has no slope.

Fraction

$$\frac{6}{x} = 6x^{-1}$$

Rewrite first:
→ $-6x^{-2}$.

What to do with $\frac{dy}{dx}$

Gradient at

$$x = 2$$

put $x = 2$

Substitute.

Gradient is 8

$$\frac{dy}{dx} = 8$$

Solve for x .

Turning point

$$\frac{dy}{dx} = 0$$

Solve, then y .

Increasing

$$\frac{dy}{dx} > 0$$

Inequality.



CALCULATOR FIRST, THEN CHECK

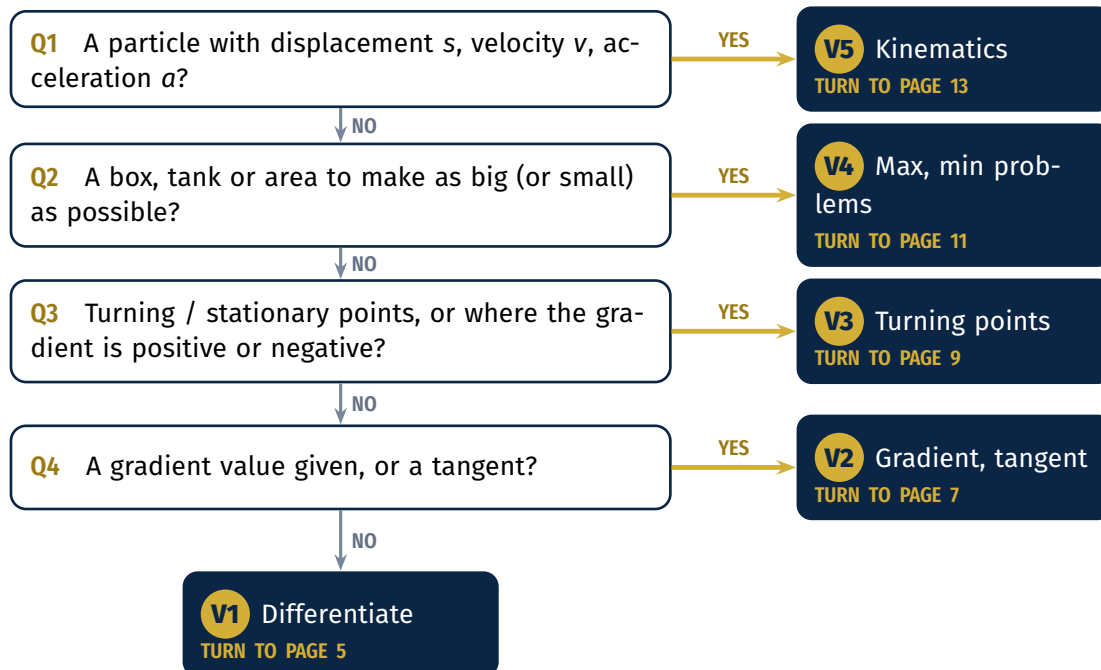
Check a derivative on the calculator: d/dx at one x value should match your formula at that x .

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Differentiate

Write every term as ax^n , then multiply by the power and reduce the power by one.

SEEN IN PAST PAPERS

4 / 29

PRACTICE QUESTIONS: 3



Your notes

Came up in: Oct/Nov 2020 P1HR Q13(a) · May 2023 P1H Q11(a) · May 2023 P1HR Q21(a) · Jun 2025 P2H Q17(a)

? What is it?**In plain words**

Each term separately:

bring the power down in front, then
take 1 off the power.

Fractions and roots first become powers: $\frac{6}{x} = 6x^{-1}$, $\sqrt{x} = x^{\frac{1}{2}}$. A number on its own differentiates to **0**.

IT LOOKS LIKE

$$5x^3 - x^2 - 6x + 4$$

$$3x^4 - \frac{6}{x} + 5$$

$$\begin{array}{c} 5x^3 \\ \downarrow \times \text{POWER, POWER} - 1 \\ 3 \times 5x^2 = 15x^2 \end{array}$$

How do I do it?**1 Rewrite**

Every term as ax^n (negative/fraction powers allowed).

2 Multiply

Coefficient $\times$ power.

3 Reduce

New power = $n - 1$.

4 Tidy

Constants vanish; keep negative powers or write as fractions.

$\times$ POWER, POWER - 1 Term by term.

V1 Differentiate

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO"Find $\frac{dy}{dx}$ "→ Term by term: $\times$ power, power -1 $\frac{16}{x}$ or $\frac{125}{t}$ in the equation→ Rewrite as $16x^{-1}$ first

A number on its own

→ Becomes 0

WORKED EXAMPLE DR ESAM

Differentiate • 3 marks

Differentiate $y = 3x^4 - \frac{6}{x} + 5$.

STEP 1 $y = 3x^4 - 6x^{-1} + 5$

◀ rewrite as powers

STEP 2 $3x^4 \rightarrow 12x^3$

◀ $\times 4$, power 3

STEP 3 $-6x^{-1} \rightarrow +6x^{-2}$

◀ $\times (-1)$, power -2

STEP 4 $5 \rightarrow 0$

◀ constant

ANSWER $\frac{dy}{dx} = 12x^3 + 6x^{-2}$

$\checkmark = 12x^3 + \frac{6}{x^2} \checkmark$

WATCH OUT

$\times -6x^{-1} \rightarrow -6x^{-2}$

$\checkmark +6x^{-2}$

Multiply by the power -1 : two negatives make a positive.**EXAM TIP**Power -1 goes to -2 (down one), not 0.**Check yourself**

calculator first • answers on page 15

1.1

May 2023 P1H

$y = 4x^3 + x^2 - 20x$. Find $\frac{dy}{dx}$.

Answer:

1.2

Practice • Medium

$y = \frac{2}{x^3}$. Find $\frac{dy}{dx}$.

Answer:

1.3

Practice • Medium

$y = \frac{4}{x}$. Find $\frac{dy}{dx}$.

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Gradient & Tangent

Gradient given: set $\frac{dy}{dx}$ equal to it and solve.

Tangent: gradient at the point, then

$$y - y_1 = m(x - x_1).$$

SEEN IN PAST PAPERS

7 / 29

PRACTICE QUESTIONS: 3



Your notes

Came up in: Oct/Nov 2020 P1HR Q13(b) • Jan 2022 P1H Q24 • May 2023 P1H Q11(b) • Nov 2023 P1H Q16 • May 2024 P1H Q18 • Nov 2024 P1H Q20 • Nov 2025 P2H Q22

? What is it?

In plain words

"The gradient is 8" means $\frac{dy}{dx} = 8$:
solve for x , then put each x into y
(not $\frac{dy}{dx}$) for the point.

A **tangent** at $x = a$: gradient m from $\frac{dy}{dx}$, point from y , then the straight-line formula.

IT LOOKS LIKE

gradient is 8

tangent at the minimum

$$y = x^3 - 40x + 1$$

$$3x^2 - 40 = 8$$

$$x = \pm 4$$

$$(4, -95), (-4, 97)$$

How do I do it?

1 Differentiate

Find $\frac{dy}{dx}$.

2 Set

 $\frac{dy}{dx}$ = the gradient (or put in x).

3 Solve

Usually a quadratic in x .

4 Back into y

Coordinates; or $y - y_1 = m(x - x_1)$ for a tangent.**SET, SOLVE, BACK INTO y** y from the curve, not from $\frac{dy}{dx}$.

V2 Gradient & Tangent

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Find the coordinates of the points at which the gradient is 8" → $\frac{dy}{dx} = 8$; then y

"Find an equation of the tangent at the minimum point" → $\frac{dy}{dx} = 0$ there: tangent $y = \text{constant}$

"A(2, -6) lies on C; the gradient at A is 16" → Two equations in a, b

WORKED EXAMPLE **PAST PAPER**

May 2024 · Paper 1H · Q18 · 5 marks

A curve C has equation $y = x^3 - 40x + 1$. Find the coordinates of both the points on C at which the gradient is 8.

STEP 1 $\frac{dy}{dx} = 3x^2 - 40$ ◀ differentiate

STEP 2 $3x^2 - 40 = 8 \Rightarrow x^2 = 16$ ◀ set and solve

STEP 3 $x = 4$ or $x = -4$ ◀ both roots

STEP 4 $y(4) = -95, \quad y(-4) = 97$ ◀ back into y

ANSWER (4, -95) and (-4, 97)✓ $3(16) - 40 = 8$ ✓**WATCH OUT**✗ only $x = 4$ ✓ $x = \pm 4$ $x^2 = 16$ has two roots.**EXAM TIP**

Write $y = \dots$ with each x substituted — the final A marks need both points.

Check yourself

calculator first · answers on page 15

2.1 May 2023 P1H

$y = 4x^3 + x^2 - 20x$. Find the x coordinates where the gradient is 4.

Answer:

2.2 Nov 2023 P1H

$y = 4x^3 - 8x + 5$. Find the x coordinates where the gradient is $\frac{1}{3}$.

Answer:

2.3 Nov 2024 P1H

$y = 2x^4 - 64x$ has a minimum point. Find the equation of the tangent there.

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM

Turning Points

Set $\frac{dy}{dx} = 0$, solve, find y . Increasing: $\frac{dy}{dx} > 0$;
decreasing: < 0 .

SEEN IN PAST PAPERS

7 / 29

PRACTICE QUESTIONS: 9



Your notes

Came up in: Oct/Nov 2020 P1H Q23 · May 2021 P2H Q21 · Jun 2022 P1HR Q22 · May 2023 P1HR Q21(b) · May 2024 P1HR Q19 · Jun 2025 P2H Q17(b)

? What is it?

In plain words

A turning (stationary) point is where the curve is **flat**: $\frac{dy}{dx} = 0$. Solve for x , then y from the curve.

Where the gradient is **positive** the curve rises; **negative**, it falls — an inequality with $\frac{dy}{dx}$.

IT LOOKS LIKE

stationary points

positive gradient

$$3x^2 - 16x - 12 = 0$$

$$(3x + 2)(x - 6) = 0$$

$$x = 6 \text{ or } x = -\frac{2}{3}$$

$$A(6, -139)$$

How do I do it?

1 Differentiate

$$\frac{dy}{dx}$$

2 = 0

Solve (factorise or formula).

3 y

Each x back into the curve.

4 Sign

Positive / negative gradient: solve $\frac{dy}{dx} > 0$ or < 0 .

FLAT = ZERO Then y from the curve.

V3 Turning Points

VARIANT 3 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

“Find the coordinates of the turning points”

→ $\frac{dy}{dx} = 0$; **y from the curve**

“the only stationary point is $(\frac{2}{3}, a)$ ”

→ $\frac{dy}{dx} = 0$ **at** $x = \frac{2}{3} \rightarrow k$, **then a**

“range of values of x for which T has a positive gradient”

→ $\frac{dy}{dx} > 0$: **quadratic inequality**

**WORKED EXAMPLE PAST PAPER**

May 2024 · Paper 1HR · Q19 · 5 marks

C: $y = x^3 - 8x^2 - 12x + 5$ has two stationary points, A and B, with the x coordinate of A greater than that of B. Find the coordinates of A.

STEP 1 $\frac{dy}{dx} = 3x^2 - 16x - 12$ ◀ differentiate

STEP 2 $(3x + 2)(x - 6) = 0$ ◀ set = 0, factorise

STEP 3 $x = 6$ (the larger) ◀ choose A

STEP 4 $y = 216 - 288 - 72 + 5 = -139$ ◀ back into y

ANSWER A(6, -139)

✓ $3(36) - 96 - 12 = 0$ ✓

**WATCH OUT**

✗ y from $\frac{dy}{dx}$ (gives 0)

✓ y from the curve

$\frac{dy}{dx}$ is the gradient, not the height.

**EXAM TIP**

Positive-gradient questions: the answer is the outside region of the quadratic (U-shape).

**Check yourself**

calculator first · answers on page 15

3.1

Jun 2025 P2H

$y = 4x^3 + 5x^2 + 2x$. Find the coordinates of the turning points.

Answer:

3.2

May 2023 P1HR

$y = x^3 - 2x^2 - 9x + 15$. For which x is the gradient positive? (3 s.f.)

Answer:

3.3

May 2021 P2H

$y = kx^2 + \frac{16}{x}$ has its only stationary point at $(\frac{2}{3}, a)$. Find a .

Answer:

V4

VARIANT 4

Maximum & Minimum Problems

Write the quantity in one letter, differentiate, set = 0, then find the maximum value.

SEEN IN PAST PAPERS

4 / 29

PRACTICE QUESTIONS: 4



Your notes

Came up in: Jan 2020 P1H Q15 · Jan 2021 P1H Q17 · Jun 2025 P2HR Q22

? What is it?**In plain words**

“Make the volume as big as possible.”

First write the

volume (or area) in terms of x only.

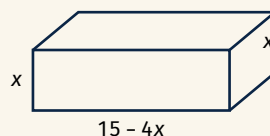
Its graph has a **top**: there $\frac{dV}{dx} = 0$.

Solve for x , then put x back into V for the **maximum value**.

IT LOOKS LIKE

$$V = x^2(15 - 4x)$$

$$y = 60x^2 - 10x^3$$



$$V = 15x^2 - 4x^3$$

$$\frac{dV}{dx} = 0$$

How do I do it?**1 Formula**

Quantity in one letter (expand it).

2 Differentiate

$$\frac{dV}{dx}$$

3 = 0

Solve; reject $x = 0$ or impossible values.

4 Value

Put x into V for the maximum.

ONE LETTER, THEN = 0

Answer the maximum value, not just x .

V4 Maximum & Minimum Problems

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Find the maximum value of V " → $\frac{dV}{dx} = 0$, then V

"There is a value of x for which the volume is a maximum. Find it" → **Stop at x (to 3 s.f.)**

" $V = \frac{1}{3}\pi y \dots$ find the maximum value of y " → **y in terms of x first**

WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 2HR · Q22 · 5 marks

A cuboid has a square end x cm by x cm and length $(15 - 4x)$ cm. Its volume is V cm³. Find the maximum value of V .

STEP 1 $V = x^2(15 - 4x) = 15x^2 - 4x^3$ ◀ one letter

STEP 2 $\frac{dV}{dx} = 30x - 12x^2$ ◀ differentiate

STEP 3 $6x(5 - 2x) = 0 \Rightarrow x = 2.5$ ◀ $x = 0$ gives no box

STEP 4 $V = 2.5^2(15 - 10) = 31.25$ ◀ maximum value

ANSWER $V_{\max} = 31.25$ cm³ ✓ $x = 2$: 28; $x = 3$: 27 — both smaller ✓

WATCH OUT

✗ answer $x = 2.5$

✓ answer $V = 31.25$

The question asks for the maximum value of V .

EXAM TIP

Check two nearby x values give smaller V — it confirms a maximum.

Check yourself

calculator first · answers on page 15

4.1 Jan 2020 P1H

$V = 15 + 16x - x^2 - 2x^3$.
Find the x that makes V
a maximum (3 s.f.).

Answer:

4.2 Jan 2021 P1H

$y = 60x^2 - 10x^3$ (from
 $V = \frac{1}{3}\pi y$). Find the
maximum value of y .

Answer:

4.3 Practice · Hard

40 m of fence makes
three sides of a pen
against a wall: width w ,
area $w(40 - 2w)$.
Maximum area?

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM**Kinematics: s, v, a** Differentiate s to get v , differentiate v to get a . At rest: $v = 0$.

SEEN IN PAST PAPERS

7 / 29

PRACTICE QUESTIONS: 3



Your notes

Came up in: Jan 2020 P1HR Q23 · Jan 2021 P1HR Q20 · Jan 2021 P2H Q19 · Jan 2022 P1HR Q23 · Jan 2022 P2H Q24 · Jun 2022 P1H Q17 · Nov 2023 P2H Q18

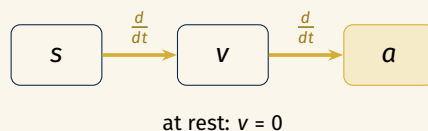
? What is it?**In plain words**

Displacement s → differentiate →
velocity v → differentiate →
acceleration a .

“Instantaneously at rest” /

“reverses direction”: $v = 0$.

“Minimum speed”: $\frac{dv}{dt} = 0$. Find t first, then the quantity asked.

**IT LOOKS LIKE**

$$s = t^3 + 5t^2 - 8t + 10$$

at rest

$$a = 5$$

How do I do it?**1 Differentiate**

$$v = \frac{ds}{dt}; \quad a = \frac{dv}{dt}.$$

2 Condition

At rest $v = 0$; given a ; min speed $\frac{dv}{dt} = 0$.

3 Solve

For t ($t \geq 0$).

4 Answer

Put t into the quantity asked (s, v or a).

$s \rightarrow v \rightarrow a$ Find t first.

V5 Kinematics: s, v, a

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Find the displacement when P is instantaneously at rest”

$$\rightarrow v = 0 \rightarrow t \rightarrow s$$

“Find the value of t when the acceleration is 5”

$$\rightarrow a = \frac{dv}{dt} = 5$$

“Find the minimum speed”

$$\rightarrow \frac{dv}{dt} = 0 \rightarrow t \rightarrow v$$

WORKED EXAMPLE PAST PAPER

January 2021 · Paper 2H · Q19 · 5 marks

$s = t^3 + 5t^2 - 8t + 10$ ($t \geq 0$). Find the displacement of P from O when P is instantaneously at rest. Give your answer as $\frac{a}{b}$.

STEP 1 $v = 3t^2 + 10t - 8$

◀ differentiate

STEP 2 $(3t - 2)(t + 4) = 0 \Rightarrow t = \frac{2}{3}$

◀ $v = 0; t \geq 0$

STEP 3 $s = \frac{8}{27} + \frac{20}{9} - \frac{16}{3} + 10$

◀ put t into s

STEP 4 $= \frac{8+60-144+270}{27} = \frac{194}{27}$

◀ common denominator

ANSWER $\frac{194}{27} \text{ m}$

$$\checkmark v\left(\frac{2}{3}\right) = \frac{4}{3} + \frac{20}{3} - 8 = 0 \checkmark$$

WATCH OUT

× putting t into v (gives 0)

✓ put t into s

The question asks for the displacement.

EXAM TIP

Reject negative t — time starts at 0.

Check yourself

calculator first · answers on page 15

5.1 Nov 2023 P2H

$s = 2t^3 - 5t^2 + 6t - 5$.
Find t when the acceleration is 5 m/s^2 .

Answer:

5.2 Jun 2022 P1H

$s = 4t^2 + \frac{125}{t}$ ($t \geq 1$).
Find the distance from O when $v = 0$.

Answer:

5.3 Jan 2021 P1HR

$s = t^3 - 9t^2 + 33t - 6$.
Find the minimum speed of P .

Answer:

CHECK YOURSELF

Answers

Your notes

Every answer below was checked with sympy (check_answers.py).

1.1 $12x^2 + 2x - 20$

1.2 $-6x^{-4}$

1.3 $-4x^{-2}$

2.1 $x = \frac{4}{3} \text{ or } -\frac{3}{2}$

2.2 $x = \pm \frac{5}{6}$

2.3 $y = -96$

3.1 $\left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$

3.2 $x < -1.19 \text{ or } x > 2.52$

3.3 $a = 36$

4.1 $x = 1.47$

4.2 320

4.3 200 m^2

5.1 $t = \frac{5}{4}$

5.2 75 m

5.3 6 m/s

THE WHOLE TOPIC ON ONE PAGE

Differentiation cheat sheet



Your notes
V1 Differentiate 4/29

$$ax^n \rightarrow anx^{n-1}$$

Rewrite fractions as powers first.

V2 Gradient 7/29

$$\frac{dy}{dx} = 8$$

Set, solve, back into y.

V3 Turning pts 7/29

$$\frac{dy}{dx} = 0$$

Then y; > 0 rising.

V4 Max / min 4/29

$$V = 15x^2 - 4x^3$$

One letter, = 0, value.

V5 Kinematics 7/29

$$s \rightarrow v \rightarrow a$$

At rest: $v = 0$.**TOP 3 LOST MARKS**

1. y found from $\frac{dy}{dx} \times$
2. one root of $x^2 = 16 \times$
3. answering x when the max value is asked $\times$

BOTTOM LINE $\times$ power, power -1 · set $\frac{dy}{dx}$ · solve · back into the original.